EMC ASSESSMENT (UKCA)

Product: SidecarTridge TOS Emulator

Model/Revision: SidecarTridge TOS Emulator Mega ST Edition / REV4.0.0

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1. PRODUCT OVERVIEW

The product is a low-voltage digital electronic device intended for use in retro computing systems. It is designed as an internal or semi-internal component (e.g. expansion board, cartridge, or interface module) and is powered by the host system.

Typical operating voltage is below 50 V DC (e.g. 3.3V / 5V / 12V).

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2. APPLICABLE REGULATIONS

This assessment is made in relation to:

- Electromagnetic Compatibility Regulations 2016 (UK SI 2016 No. 1091)

The product does not fall under:

- Electrical Equipment (Safety) Regulations 2016
  (as it operates below 50 V DC)

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3. DESIGN CHARACTERISTICS RELEVANT TO EMC

- The device contains standard digital logic components (e.g. CMOS/TTL, microcontrollers, memory devices).
- No radio transmitter or intentional RF radiator is present.
- The device does not include wireless communication functionality.
- The design does not incorporate high-frequency switching power supplies (unless provided externally by the host system).
- Signal frequencies are limited to those typical of legacy computing platforms (generally low MHz range or below).
- External connections are limited in length and are typically internal to the host system or via short cables.
- The device includes a microcontroller-based system operating at moderate clock frequencies typical of embedded systems.
- A USB interface is present for local data transfer; it is not intended for continuous high-speed communication during normal operation.

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4. EMISSION ASSESSMENT

Due to the design characteristics described above:

- The product is not expected to generate significant electromagnetic emissions.
- Any emissions are comparable to those of standard digital logic circuits used in legacy computing systems.
- The absence of high-frequency switching regulators and RF transmitters significantly reduces emission risk.
- The USB interface is used intermittently for file transfer and does not represent a continuous high-frequency emission source.

Conclusion:
The product presents a low risk of causing electromagnetic interference when used as intended.

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5. IMMUNITY ASSESSMENT

- The product is designed to operate within the electromagnetic environment provided by the host system.
- Immunity performance depends on the overall system in which the product is installed.
- The product does not include specific shielding or filtering beyond standard design practices.

Conclusion:
The product is expected to operate correctly in typical residential, commercial, and light-industrial environments when integrated into a compliant host system.

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6. INSTALLATION AND USE CONDITIONS

- The product is intended for integration into a larger electronic system.
- It is not designed to operate as a standalone end-user device.
- Proper operation and EMC performance depend on correct installation within a compliant host system.

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7. OVERALL CONCLUSION

Based on the design characteristics and intended use, the product is considered to comply with the essential requirements of the Electromagnetic Compatibility Regulations 2016, provided that it is installed and used in accordance with its intended purpose.

This assessment is supported by the use of designated standards such as:

- BS EN IEC 61000-6-1:2019 (Immunity)
- BS EN IEC 61000-6-3:2021 (Emissions)

No additional EMC mitigation measures are required based on the design and intended use.

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8. RESPONSIBLE PERSON

Name: Diego Parrilla Santamaria
Position: CEO
Company: GOODDATA LABS SL

Date: 23 April 2026

Signature: